

Rithvik Allamaneni

allamaneni.1@buckeyemail.osu.edu | 513-525-3319 | <https://github.com/Allam235> | Cincinnati, Ohio

EDUCATION

The Ohio State University, Columbus, OH Expected Graduation Date: May 2028
Bachelor of Science Honors in Computer Science & Engineering GPA(4.00 scale): 3.83
Specialization: Database Systems and Artificial Intelligence; Minor: Biomedical Engineering

QUALIFICATIONS

- **Programming Languages:** Java, Maven, Ruby, Rails, Node.js, React.js, C, x86 assembly, Python
- **Technologies:** DBT, Snowflake, Prefect, SQL Server, Linux CLI, Eclipse, VSCode, Git, GitHub, OnShape, Agile
- **Relevant Coursework:** Operating Systems, Web Application Development, Java 1 & 2, Data Structures & Algorithms, Programming in C and Assembly, Linear Algebra, Probability & Statistics

WORK EXPERIENCE

Medpace

Data Engineering Intern May 2026 - August 2026

- Migrated enterprise data warehouse pipelines from **SQL Server/SSIS** to **Snowflake** by rebuilding legacy transformation logic in **dbt** and **Prefect**, preserving functional parity while modernizing ETL workflows
- Developed and deployed data products for clinical monitoring applications by translating business requirements into dimensional models, snapshots, and analytical views that aligned with the medallion data architecture
- Engineered a scalable migration strategy for a 175M+ row legacy snapshot table by implementing SCD Type II with delta detection and incremental loading, reducing migration costs by **80%** while preserving historical accuracy
- Resolved production data integrity issues by developing row-hash validation workflows to compare environments, identify transformation discrepancies, and restore broken fact-to-dimension relationships before deployment

The Ohio State University

Teaching Assistant for CSE 2421: Programming in C and Assembly Columbus, OH January 2026 – Present

- Supported 40+ students during office hours, guiding debugging C programs involving pointers & dynamic memory.
- Evaluated and graded programming assignments in C and x86 assembly, reinforcing concepts in memory safety, computer architecture, and performance

PROJECT EXPERIENCE

College Housing Match Platform — Team Lead Housing January 2026 – May 2026

- Led an **Agile** based development team in architecting and building the housing portal and data scraping for a full-stack roommate matching platform using **Next.js**, **React**, and **Tailwind**, by translating product requirements into structured feature tickets and user stories while coordinating sprint-based execution
- Designed and implemented backend infrastructure with **Supabase (PostgreSQL, Auth, API)** and **Next.js**, modeling relational schemas for housing listings, user swipes, matches, and roommate preference data
- Strengthened my ability to break large product goals into scoped, deliverable features, distribute tasks based on team strengths, and coordinate parallel development across authentication, discovery, chat, and housing modules

Hack AI 2026 Ohio State University – 2nd Place, Scientific MOOD–FAIR Challenge Track February 22nd 2026

- Built a probabilistic deep learning model in PyTorch using an ImageNet-pretrained ResNet18 (TorchVision) backbone to predict multi-scale drought indices from beetle imagery, producing calibrated mean and variance outputs
- Trained the model using Gaussian negative log-likelihood (NLL) loss and incorporated domain priors with NumPy-based computation to improve generalization performance
- Strengthened my ability to design reliable ML systems, gaining practical experience in uncertainty modeling, model calibration, and improving performance under distribution shifts

INVOLVEMENT

- Squash Club, Team Member (Fall 2025 – Present)
- Collaborative Software Development Club, Web Design Team Leader (Fall 2025 – Present)
- Buckeye Formula Club, Telemetry Member (Fall 2025 – Present)